

10_FORMAL_METHODS.md

TASK 10

FORMAL METHODS MODELLING & VERIFICATION

Status

Mandatory

Owner

Formal Methods Engineer

Supporting Guides

Project Director

Requirements Engineer

Security Architect

Solution Architect

Backend Lead

QA & Security Lead

Principal Engineer

Estimated Time

2 to 3 Days

Dependencies

00_PROJECT_CONTEXT.md

00_ENGINEERING_RULES.md

00_PROJECT_MEMORY.md

10_FORMAL_METHODS_ENGINEER.md

02_REQUIREMENTS.md

03_ARCHITECTURE.md

06_BACKEND.md

09_TESTING.md

Mission

Create formal behavioural models for the Secure Dance Academy Management System and verify that critical workflows satisfy correctness, safety and security properties.

The objective is to provide mathematical evidence that the implemented system behaves as intended.

Objectives

Model critical workflows.

Create Finite State Machines.

Convert FSMs into Petri Nets.

Verify correctness.

Verify reachability.

Verify deadlock freedom.

Verify safety.

Verify liveness.

Produce professional documentation.

Formal Methods Standards

Follow accepted modelling practices.

Use deterministic state transitions where appropriate.

Maintain one-to-one traceability between requirements, models and implementation.

Every model must have a documented purpose.

Workflow Selection

Model the most critical system behaviours.

Authentication

Password Reset

Attendance Recording

Attendance Approval

Medical Record Access

Performance Recording

Role Assignment

User Registration

Session Management

Administrative Approval

Prioritize workflows that affect security or data integrity.

Finite State Machine Design

Create complete FSMs.

Every FSM must include

State Definitions

Initial State

Final State

Events

Transitions

Guards

Error States

Recovery States

State Descriptions

Transition Table

Avoid unnecessary states.

State Design

Use meaningful business states.

Examples

Logged Out

Authenticating

Authenticated

Session Expired

Attendance Pending

Attendance Confirmed

Attendance Rejected

Performance Submitted

Performance Approved

Medical Record Updated

Do not model implementation details.

Model business behaviour.

Transition Design

Every transition should define

Trigger

Source State

Target State

Guard Condition

Expected Result

Security Requirement

Document invalid transitions.

Transition Tables

Create detailed transition tables.

Include

Current State

Event

Condition

Next State

Output

Security Rule

Every transition should be traceable.

Petri Net Modelling

Convert every FSM into a Petri Net.

Document

Places

Transitions

Tokens

Arcs

Initial Marking

Final Marking

Maintain consistency with the FSM.

Reachability Analysis

Verify

All required states are reachable.

Invalid states remain unreachable.

No isolated states exist.

Every workflow reaches completion.

Document the reachability graph.

Deadlock Analysis

Verify

No workflow becomes permanently blocked.

Resources are released correctly.

Users are never trapped in invalid states.

If deadlocks exist

Explain the cause.

Recommend architectural improvements.

Safety Verification

Prove

Unauthorized transitions cannot occur.

Restricted resources remain protected.

Business rules are enforced.

Medical records remain protected.

Invalid workflows terminate safely.

Document every safety property.

Liveness Verification

Verify

Valid workflows always complete.

Users eventually reach completion.

Sessions terminate correctly.

Resources are eventually released.

No infinite waiting occurs.

Security Verification

Model

Authentication

Authorization

RBAC

Session Management

Password Reset

Medical Record Access

Administrative Approval

Verify that security controls prevent unauthorized state changes.

Traceability

Link every model to

Requirement ID

Use Case

Business Rule

API

Database Entity

Test Case

Security Requirement

Maintain complete traceability.

Tooling

Recommended tools

PIPE

WoPeD

Draw.io

PlantUML

Visual Paradigm

Graphviz

Use consistent notation throughout the project.

Documentation

Produce

FSM Diagrams

Petri Net Diagrams

State Definitions

Transition Tables

Reachability Analysis

Deadlock Analysis

Safety Analysis

Liveness Analysis

Formal Verification Report

Traceability Matrix

Diagram Source Files

Every diagram should include interpretation.

Deliverables

Formal Specification

Finite State Machines

Petri Nets

Reachability Graphs

Deadlock Analysis

Safety Proof

Liveness Proof

Verification Report

Formal Methods Appendix

Supporting Diagrams

Acceptance Criteria

Critical workflows are modelled.

FSMs are complete.

Petri Nets are complete.

Reachability has been verified.

Deadlocks have been analysed.

Safety properties have been demonstrated.

Liveness properties have been demonstrated.

Traceability is complete.

Documentation is complete.

Update Project Memory

Mark

Formal Methods

Completed

Record

Models Created

Verified Workflows

Outstanding Limitations

Tool Versions

Verification Results

Next Task

Documentation & Final Report

Formal Review Checklist

Verify

Models match requirements.

FSMs are internally consistent.

Petri Nets match FSMs.

Reachability is correct.

No unexpected deadlocks exist.

Safety properties are satisfied.

Liveness properties are satisfied.

Documentation is complete.

Traceability is complete.

Definition of Done

The Formal Methods deliverables provide clear mathematical evidence that critical workflows behave correctly and securely.

Every selected workflow has been modelled, verified and documented.

The completed analysis demonstrates postgraduate understanding of software verification and satisfies the Formal Methods assessment requirements while supporting the overall security and correctness of the Secure Dance Academy Management System.